DAY — 12 SEAT NUMBER

2011	III	14	1100	J - 601	(E)

CHEMISTRY PAPER - II (55)
(ORGANIC AND INORGANIC CHEMISTRY)

Time : 2 Hrs. (4 Pages) Max. Marks : 40

Note : (i) *All questions carry equal marks.*

(ii) *Give balanced equations and draw neat and labelled diagrams wherever necessary.*

(iii) *Figures to the right hand side indicate full marks.*

(iv) *Answer to every new question must be started on a new page.*

Q. 1. Select and write the most appropriate answer from the given alternatives for each sub-question : [8]

(i) Valium is a/an (1)

(a) Analgesic (b) Tranquilizer

(c) Antibiotic (d) Antacid

(ii) In Actinides the last electron enters which sub-shell? (1)

(a) 4f (b) 5f

(c) 6d (d) 7s

(iii) Ethylidene dichloride when boiled with aqueous NaOH gives (1)

(a) Formaldehyde (b) Acetone

(c) Butanone (d) Acetaldehyde

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(iv) Phenol on treatment with concentrated H_2SO_4 at room temperature gives (1)

(a) Ortho phenol sulphonic acid

(b) Para phenol sulphonic acid

(c) Meta phenol sulphonic acid

(d) Benzene sulphonic acid

(v) Propanone when reacted with Zn-Hg|conc. HCl produces (1)

(a) ethane (b) n-propyl alcohol

(c) propane (d) iso-propyl alcohol

(vi) $CH_3COOH + PCl_5 \xrightarrow{\Delta} A + POCl_3 + HCl$

$A + C_2H_5OH \xrightarrow[\text{Pyridine}]{\Delta} B + HCl$

The product 'B' is : (1)

(a) Ethyl acetate (b) Methyl acetate

(c) Ethyl formate (d) Methyl formate

(vii) Number of possible structures for the amines having molecular formula C_3H_9N are (1)

(a) 2 (b) 3

(c) 4 (d) 5

(viii) Stachyose is (1)

(a) Monosaccharide (b) Disaccharide

(c) Trisaccharide (d) Tetrasaccharide

Q. 2. (A) Attempt any ONE : [8]

(i) What are amines? Give IUPAC name of ethyl methyl amine. (2)

(ii) Give chemical reactions involved in the preparation of Terylene. (2)

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(B) Attempt any ONE :

(i) How is ethyl methyl ether prepared from

(a) diazomethane

(b) sodium ethoxide? (2)

(ii) What are Triglycerides? Give 'one example each' of fat and oil. (2)

(C) Answer the following :

(i) Explain the cause of lanthanide contraction. (2)

(ii) Differentiate between carbonium ion and carbanion. (2)

Q. 3. (A) Attempt any ONE : **[8]**

(i) How will you obtain (3)

(a) diacetonamine from acetone?

(b) urotropine from formaldehyde?

(c) acetaldoxime from acetaldehyde?

(ii) Define : (3)

(a) Optical activity

(b) Asymmetric carbon atom.

Draw structures of 'two enantiomers' of lactic acid.

(B) Attempt any ONE :

(i) What is the action of excess of phenyl hydrazine on glucose? (3)

(ii) How is carbolic acid prepared from cumene? Give 'two uses' of phenol. (3)

(C) Answer the following :

Define : (i) Fibre and (ii) Ether. (2)

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Q. 4. (A) **Answer the following :** [8]

Explain :

(i) Electromeric effect

(ii) Homolytic fission (4)

(B) Attempt any ONE :

(i) What is the action of excess of methyl magnesium iodide on ethyl ethanoate? (4)

(ii) (a) What is the action of the following on ethanol : (2)

(i) $K_2Cr_2O_7$ and dil. H_2SO_4.

(ii) Thionyl chloride

(b) Boiling points of alcohols are much higher than those of the corresponding alkanes - Explain. (2)

Q. 5. (A) **Attempt any ONE :** [8]

(i) What are aldehydes and ketones? Write a note on Cannizzaro reaction. (4)

(ii) How is iodoform prepared from ethanol? (4)

(B) Attempt any TWO :

(i) What are disinfectants? (2)

Give structure of saccharin.

(ii) How will you convert : (2)

(a) $-C \equiv N$ into $-CH_2-NH_2$

(b) $-NH_2$ into $-OH$

(iii) Name 'two elements' that form a pair of chemical twins. Explain, why are those elements known as chemical twins. (2)

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